

evaluation, we repeat the experiment over five independent runs with different random samplings of the training data. In each run, we randomly sample 512 human-written/LLM-generated text pairs to form the training set, while using a fixed, disjoint test set of 1,000 pairs. We further evaluate on the recent OpAI-Bench [69], which contains progressive human-to-AI revision trajectories across four domains and includes texts revised by more recent generators, including *GPT-5.4*, *GPT-5.4-nano*, *Gemini-2.5-Flash*, and *Qwen3-8B*. Full experimental details and results are provided in Appendix F.8.

Baselines. In our primary DetectRL evaluation, we compare S2D against thirteen baselines across train-free and train-based categories. The train-free baselines include: *Likelihood* [70], Log Rank Ratio (*LRR*) [70], Fast-DetectGPT (*FDGPT*) [20], *Binoculars* [21], Implicit Reward Model (*IRM*) [71], *Lastde++* [72], *SpecDetect++* [73], and Normalized Temperature Sensitivity (*NTS*) [74]. The train-based baselines consist of: *RoBERTa-Large* [25], *RAIDAR* [22], Imitate Before Detection (*ImBD*) [75], *RepreGuard* [33], and Learn-to-Distance (*L2D*) [23]. We further conduct controlled comparisons against logistic regression, a linear SVM, and a small MLP trained on the same pooled observer representations without steering, to assess whether a standard discriminative readout alone can account for the observed performance. The complete comparisons are reported in Appendix F.8.

Evaluation Metrics. We evaluate detection performance using AUROC and the true positive rate (TPR) at two fixed false positive rate (FPR) levels, namely TPR@1% and TPR@0.01%⁵. These metrics capture both overall discrimination and performance in the high-precision regime where false positives must be tightly controlled. Further experimental details are provided in Appendix D.

4.1 Detection Performance and Robustness

Detection under ID and OOD settings. We first evaluate detection performance under both in-distribution (ID) and out-of-distribution (OOD) settings, as summarized in Table 2. The detector is trained on text generated by a specific source model and evaluated either on the same generator (ID, diagonal entries) or on different generators (OOD, off-diagonal entries).

Under the ID setting, S2D performs strongly across all generators when the training and test distributions are aligned. RepreGuard and L2D are the most competitive baselines, often achieving TPRs above 99% at low FPR, while other training-based methods are less consistent across generators. L2D relies on rewriting, whereas S2D uses a single forward pass of the frozen observer model at inference time; efficiency comparisons are provided in Appendix F.6. In the OOD setting, performance gaps become more pronounced, and S2D remains competitive against most baselines.

Robustness to Adversarial Attacks. We then evaluate robustness under adversarial manipulations, including three paraphrasing attacks (i.e., *Polish*, *Back Translation*, *DIPPER*) and three perturbation attacks (i.e., *Character*, *Word*, and *Character+Word*), following the previous setup [20]. As shown in Table 3, paraphrasing attacks substantially degrade all detectors, highlighting the challenge of semantic-preserving rewrites. Nevertheless, S2D is highly competitive and often achieves the best or near-best performance, whereas baseline methods experience severe degradation. This robustness stems from S2D’s representation reshaping, which enforces a more discriminative feature space and preserves separability even under attacks. In contrast, perturbation-based attacks have a much milder impact, as they primarily introduce local or surface-level noise.

Robustness to Text Length. Our main test set has an average length of approximately 267 tokens, with 95% of samples shorter than 435 tokens. Figure 2a shows that detection generally benefits from additional context within this range. We further examine substantially longer sequences, up to 4096 tokens, using a separate diagnostic set. Performance continues to improve with context length before plateauing, while the score overlap between human and LLM-generated text remains low. In particular, we observe no evidence that repeatedly injecting the same steering vector causes representation collapse or performance degradation within the evaluated range. Full long-context results are reported in Appendix F.4.

Comparison with Watermarking Methods. An interesting question is whether S2D can outperform existing watermarking methods. Although this comparison is inherently unfair—since

⁵Since the DetectRL test sets contain 1,000 human-written samples, TPR@0.01% is below the empirical FPR resolution and is obtained by linear interpolation of the ROC curve. We further examine the accuracy of this interpolation using a substantially larger null set in Appendix F.2.

Train ↓	Detector ↓	Test: ChatGPT			Test: Llama-2-70B			Test: Google-PaLM			Test: Claude-Instant		
		AUROC	TPR@1%	TPR@.01%	AUROC	TPR@1%	TPR@.01%	AUROC	TPR@1%	TPR@.01%	AUROC	TPR@1%	TPR@.01%
Train-free Methods													
N/A	Likelihood	89.91	73.00	0.40	89.32	80.30	5.80	85.73	69.30	13.10	76.72	5.50	0.40
	LRR	88.95	78.00	0.01	89.51	81.50	38.80	85.66	73.70	21.30	76.75	7.00	0.20
	FDGPT	94.70	51.90	0.06	96.85	70.10	32.90	95.26	66.00	50.40	58.37	4.10	0.70
	Binoculars	94.71	90.67	87.00	99.15	96.70	88.90	94.50	91.40	89.30	87.70	37.50	13.70
	IRM	99.08	90.70	83.10	99.20	98.20	95.50	99.11	91.30	85.20	95.50	69.20	35.10
	Lastde++	94.32	53.10	4.27	96.07	65.77	20.30	94.85	63.07	47.80	69.38	10.23	2.33
	SpecDetect++	94.19	38.90	3.07	94.76	54.33	2.60	89.46	42.30	15.07	69.90	5.70	0.40
NTS	93.48	56.60	6.00	96.93	76.40	57.30	94.75	71.60	36.80	66.72	8.90	2.30	
Train-based Methods													
ChatGPT	RoBERTa-L	98.99 $\pm$.12	98.90 $\pm$.25	98.10 $\pm$.45	98.82 $\pm$.15	63.70 $\pm$.15	44.80 $\pm$.21	98.08 $\pm$.22	45.60 $\pm$.20	25.40 $\pm$.25	98.95 $\pm$.18	76.90 $\pm$.16	15.80 $\pm$.15
	RAIDAR	99.89 $\pm$.05	99.90 $\pm$.08	99.10 $\pm$.15	98.49 $\pm$.21	92.20 $\pm$.85	32.60 $\pm$.18	94.53 $\pm$.60	75.40 $\pm$.14	51.90 $\pm$.19	87.40 $\pm$.11	67.80 $\pm$.19	0.10 $\pm$.05
	ImBD	99.94 $\pm$.04	99.90 $\pm$.06	98.72 $\pm$.20	99.98 $\pm$.02	99.90 $\pm$.05	93.64 $\pm$.90	99.65 $\pm$.10	95.30 $\pm$.65	84.82 $\pm$.11	86.18 $\pm$.13	25.20 $\pm$.24	9.34 $\pm$.80
	L2D	99.99 $\pm$.01	100.0 $\pm$.00	99.90 $\pm$.05	99.80 $\pm$.11	99.50 $\pm$.12	95.50 $\pm$.70	86.55 $\pm$.12	57.80 $\pm$.18	31.90 $\pm$.22	83.80 $\pm$.21	40.10 $\pm$.45	12.30 $\pm$.24
	RepreGuard	99.84 $\pm$.10	99.80 $\pm$.00	99.70 $\pm$.00	99.54 $\pm$.12	99.43 $\pm$.04	99.40 $\pm$.00	97.19 $\pm$.12	81.40 $\pm$.21	79.70 $\pm$.41	98.25 $\pm$.16	78.53 $\pm$.11	70.40 $\pm$.90
	S2D (ours)	99.99 $\pm$.00	100.0 $\pm$.00	99.90 $\pm$.00	100.0 $\pm$.00	99.95 $\pm$.00	99.90 $\pm$.05	98.93 $\pm$.25	91.93 $\pm$.12	90.83 $\pm$.11	98.31 $\pm$.34	83.02 $\pm$.38	79.48 $\pm$.48
Llama-2	RoBERTa-L	98.97 $\pm$.20	98.90 $\pm$.30	98.10 $\pm$.40	99.90 $\pm$.05	99.90 $\pm$.04	97.10 $\pm$.60	99.82 $\pm$.06	96.80 $\pm$.50	83.50 $\pm$.12	99.11 $\pm$.15	90.70 $\pm$.80	57.90 $\pm$.22
	RAIDAR	99.98 $\pm$.02	100.0 $\pm$.00	92.40 $\pm$.12	99.58 $\pm$.12	96.80 $\pm$.50	70.80 $\pm$.15	97.56 $\pm$.40	84.10 $\pm$.11	61.30 $\pm$.18	88.58 $\pm$.10	43.70 $\pm$.20	6.30 $\pm$.80
	ImBD	99.99 $\pm$.01	99.90 $\pm$.05	99.80 $\pm$.05	99.99 $\pm$.01	99.90 $\pm$.04	99.70 $\pm$.05	99.89 $\pm$.04	97.50 $\pm$.40	94.51 $\pm$.50	94.39 $\pm$.60	49.70 $\pm$.18	27.36 $\pm$.19
	L2D	99.98 $\pm$.02	99.90 $\pm$.05	95.60 $\pm$.80	99.80 $\pm$.08	99.30 $\pm$.15	77.60 $\pm$.12	93.82 $\pm$.90	77.40 $\pm$.13	51.40 $\pm$.21	94.47 $\pm$.55	70.80 $\pm$.15	2.40 $\pm$.50
	RepreGuard	99.94 $\pm$.00	99.78 $\pm$.04	99.62 $\pm$.08	99.94 $\pm$.01	99.40 $\pm$.00	99.35 $\pm$.09	97.35 $\pm$.14	82.85 $\pm$.52	81.32 $\pm$.78	98.41 $\pm$.20	81.05 $\pm$.50	74.28 $\pm$.26
	S2D (ours)	100.0 $\pm$.00	100.0 $\pm$.00	99.90 $\pm$.00	99.99 $\pm$.00	99.90 $\pm$.00	99.80 $\pm$.00	99.48 $\pm$.09	95.42 $\pm$.41	94.50 $\pm$.56	99.36 $\pm$.10	91.20 $\pm$.65	88.63 $\pm$.95
PaLM	RoBERTa-L	99.99 $\pm$.01	99.90 $\pm$.03	99.30 $\pm$.15	99.97 $\pm$.01	99.90 $\pm$.04	80.20 $\pm$.15	99.98 $\pm$.01	99.60 $\pm$.10	99.50 $\pm$.15	99.70 $\pm$.10	99.70 $\pm$.08	88.70 $\pm$.11
	RAIDAR	99.96 $\pm$.02	99.90 $\pm$.03	71.10 $\pm$.18	99.96 $\pm$.02	99.90 $\pm$.04	60.90 $\pm$.20	99.97 $\pm$.02	99.40 $\pm$.15	93.00 $\pm$.80	99.71 $\pm$.08	98.10 $\pm$.30	11.30 $\pm$.15
	ImBD	99.85 $\pm$.05	96.45 $\pm$.50	85.53 $\pm$.11	99.93 $\pm$.03	98.20 $\pm$.25	95.32 $\pm$.80	99.78 $\pm$.08	96.00 $\pm$.50	88.01 $\pm$.11	93.67 $\pm$.70	44.20 $\pm$.18	25.12 $\pm$.21
	L2D	100.0 $\pm$.00	100.0 $\pm$.00	100.0 $\pm$.00	99.92 $\pm$.03	100.0 $\pm$.00	24.50 $\pm$.25	99.98 $\pm$.01	99.70 $\pm$.08	95.70 $\pm$.60	99.98 $\pm$.01	99.90 $\pm$.03	97.70 $\pm$.40
	RepreGuard	99.92 $\pm$.00	99.58 $\pm$.13	99.53 $\pm$.04	99.94 $\pm$.00	99.62 $\pm$.11	99.50 $\pm$.12	96.95 $\pm$.17	85.38 $\pm$.80	84.08 $\pm$.81	96.28 $\pm$.50	76.97 $\pm$.22	72.40 $\pm$.20
	S2D (ours)	100.0 $\pm$.00	100.0 $\pm$.00	100.0 $\pm$.00	99.99 $\pm$.00	100.0 $\pm$.00	99.60 $\pm$.00	99.88 $\pm$.02	98.25 $\pm$.40	97.60 $\pm$.52	99.74 $\pm$.11	94.52 $\pm$.16	92.60 $\pm$.18
Claude	RoBERTa-L	80.91 $\pm$.15	14.00 $\pm$.12	9.70 $\pm$.80	80.26 $\pm$.14	18.20 $\pm$.16	16.30 $\pm$.15	74.82 $\pm$.18	26.95 $\pm$.18	25.00 $\pm$.20	99.97 $\pm$.01	99.80 $\pm$.05	99.60 $\pm$.10
	RAIDAR	99.91 $\pm$.03	99.70 $\pm$.10	46.70 $\pm$.22	97.32 $\pm$.50	88.90 $\pm$.12	56.20 $\pm$.20	93.60 $\pm$.80	73.50 $\pm$.14	20.20 $\pm$.15	99.98 $\pm$.01	99.90 $\pm$.04	94.70 $\pm$.80
	ImBD	99.99 $\pm$.01	99.90 $\pm$.04	99.71 $\pm$.10	99.97 $\pm$.02	99.90 $\pm$.04	84.31 $\pm$.15	99.88 $\pm$.04	97.40 $\pm$.35	91.60 $\pm$.80	99.05 $\pm$.15	86.70 $\pm$.11	60.14 $\pm$.18
	L2D	99.92 $\pm$.02	99.90 $\pm$.04	82.20 $\pm$.15	99.90 $\pm$.05	99.80 $\pm$.08	99.10 $\pm$.20	97.88 $\pm$.40	92.20 $\pm$.80	79.80 $\pm$.13	99.96 $\pm$.02	99.90 $\pm$.04	97.60 $\pm$.40
	RepreGuard	99.92 $\pm$.00	99.47 $\pm$.11	99.18 $\pm$.19	99.89 $\pm$.01	98.82 $\pm$.26	98.10 $\pm$.19	95.08 $\pm$.27	70.38 $\pm$.93	68.45 $\pm$.54	98.42 $\pm$.26	85.85 $\pm$.62	81.47 $\pm$.20
	S2D (ours)	99.98 $\pm$.03	99.98 $\pm$.04	99.93 $\pm$.04	99.99 $\pm$.01	100.0 $\pm$.00	99.98 $\pm$.04	98.92 $\pm$.20	92.58 $\pm$.69	89.45 $\pm$.16	99.99 $\pm$.00	99.95 $\pm$.09	99.95 $\pm$.09

Table 2: **In-distribution and out-of-distribution performance comparison.** Methods are categorized into train-free and train-based settings. Our proposed S2D demonstrates robust generalization, achieving state-of-the-art or highly competitive results across the majority of evaluated scenarios. Best and second-best results are highlighted in **darker** and **lighter** blue, respectively.

Detector ↓	No Attack		Paraphrasing Attacks						Perturbations					
			Polish		Back Trans.		DIPPER		Character		Word		Char + Word	
	AUROC	TPR@1%	AUROC	TPR@1%	AUROC	TPR@1%	AUROC	TPR@1%	AUROC	TPR@1%	AUROC	TPR@1%	AUROC	TPR@1%
Binoculars	95.40	62.80	73.54 (-22.9%)	0.53 (-99.2%)	90.93 (-4.7%)	53.73 (-14.4%)	93.67 (-1.8%)	58.14 (-7.4%)	94.76 (-0.7%)	57.73 (-8.1%)	94.23 (-1.2%)	58.73 (-6.5%)	95.01 (-0.4%)	61.07 (-2.8%)
L2D	95.80	88.33	56.14 (-41.4%)	1.20 (-98.6%)	92.36 (-3.6%)	15.73 (-82.2%)	87.72 (-8.4%)	3.20 (-96.4%)	83.43 (-12.9%)	15.20 (-82.8%)	95.70 (-0.1%)	80.13 (-9.3%)	94.21 (-1.7%)	42.13 (-52.3%)
RepreGuard	97.02	85.30	83.88 (-13.5%)	10.53 (-87.7%)	91.67 (-5.5%)	45.33 (-46.9%)	94.23 (-1.9%)	73.67 (-13.6%)	96.32 (-0.7%)	83.87 (-1.7%)	92.88 (-4.3%)	78.67 (-7.8%)	94.29 (-2.8%)	85.00 (-0.4%)
S2D (ours)	98.87	96.68	82.06 (-17.0%)	23.60 (-75.6%)	93.45 (-5.5%)	67.33 (-30.4%)	98.66 (-0.2%)	84.91 (-12.2%)	97.76 (-1.1%)	90.53 (-6.4%)	96.73 (-2.2%)	90.87 (-6.0%)	95.77 (-3.1%)	91.47 (-5.4%)

Table 3: **Robustness evaluation.** “No Attack” denotes performance on original text, and values in parentheses indicate the relative percentage change $((\text{Attack} - \text{No attack}) / \text{No attack} \times 100\%)$ compared to this baseline (positive values indicate performance gains under attack). Best and second-best results are highlighted in **darker** and **lighter** blue, respectively.

watermarking methods actively embed detectable signals while S2D is a passive detector—it helps position S2D relative to these approaches.⁶

Table 4 compares S2D with three watermarking methods, namely Gumbel-max [76], Green-red [11], and SynthID [77], using their default detectors, which have full knowledge of the watermarking mechanisms. Interestingly, S2D achieves stronger performance at lower temperature ($T = 0.3$), where watermark signals are weaker due to reduced generation entropy. In contrast, watermarking methods perform better at higher temperatures, where the embedded signals are more pronounced. The performance of S2D in this setting remains reasonable: even without knowledge of the watermarking scheme, it still captures distributional artifacts induced by the watermarking generation process. These artifacts are more pronounced in low-entropy settings, where watermark signals are weak, while at higher entropy levels, stronger embedded watermark signals favor watermark-specific detectors.

4.2 Ablation Study

Importance of Steering. We assess the role of steering by comparing S2D with a variant without steering, which does not incorporate the steering vector and instead estimates the vMF parameters

⁶To reflect a realistic setting, S2D is trained once on human-written and non-watermarked LLM-generated text, without retraining for specific watermarking schemes, and is directly applied at evaluation.

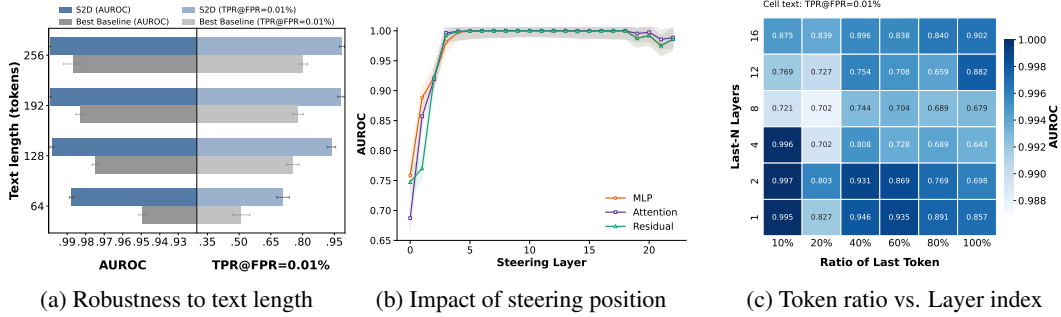


Figure 2: **Analysis of S2D performance.** (a) Comparison of detection stability across varying input lengths. (b) Detection performance across steering layers, showing that intermediate layers consistently achieve the best performance. (c) Performance heatmap as a function of last-token selection ratio and the number of aggregated layers.

Temp.	Method	64 tokens			128 tokens			256 tokens		
		Gumbel	Green-red	SynthID	Gumbel	Green-red	SynthID	Gumbel	Green-red	SynthID
		AUROC / TPR@1%FPR (%)								
0.3	Watermark Det. [†]	99.9 / 98.0	82.0 / 12.0	81.8 / 15.9	99.9 / 99.7	99.6 / 95.3	84.8 / 28.0	99.9 / 99.9	99.8 / 99.4	84.5 / 34.0
	S2D (ours)	77.6 / 20.0	96.7 / 60.7	85.1 / 26.0	96.6 / 62.8	98.4 / 81.3	96.5 / 69.0	100 / 98.5	99.9 / 98.6	99.9 / 98.1
0.7	Watermark Det. [†]	99.9 / 97.0	80.2 / 9.3	99.3 / 89.8	99.9 / 99.8	99.6 / 95.7	99.7 / 95.8	99.9 / 99.9	100 / 99.8	99.5 / 97.5
	S2D (ours)	65.4 / 12.3	90.5 / 33.0	70.0 / 10.9	89.3 / 44.7	89.0 / 41.7	84.2 / 32.7	99.6 / 91.7	95.9 / 70.4	97.6 / 79.0

[†] The defaulted detectors provided by the original watermarking algorithms.

Table 4: **Detection performance across different text lengths and temperatures.** Comparison with default watermark detectors. **Bold** indicates better performance within each setting.

directly from training representations. As shown in Figure 4, S2D w/o steering exhibits substantially weaker separation between human- and LLM-generated texts, with significant overlap between the two distributions. This highlights the importance of incorporating the steering vector.

Steering Position. We first analyze the impact of the steering position. As shown in Figure 2b, for the default *Llama-3.1-8B* observer, intermediate-layer injection performs substantially better than very early injection, while the exact insertion point within a Transformer block (i.e., residual stream, attention output, or MLP output) has a much smaller effect.

We further examine how this preferred region changes with model depth by performing steering-layer sweeps on *Llama-3.2-1B*, *Meta-Llama-3-8B*, and *Meta-Llama-3-70B*. Across all three scales, strong performance is attained over a range of intermediate depths rather than at a single isolated layer. However, the best-performing layer does not follow a fixed proportional scaling law with network depth. These results suggest that intermediate depth is a useful search region, while the exact steering layer remains model dependent. Full cross-scale results are provided in Appendix F.5.

Token Selection and Layer Aggregation. We analyze the effect of the last-token ratio K and the number of aggregated layers N (introduced in Section 2.2). As shown in Figure 2c, performance exhibits a non-monotonic pattern across both factors. Strong performance is observed in two regimes: the bottom-left region, which uses a small fraction of tokens from a few top layers, and the top-right region, where deeper aggregation is combined with larger token ratios. In contrast, intermediate configurations perform worse. This pattern suggests a trade-off between signal concentration and aggregation. Detection-relevant signals are concentrated in the final layers and later tokens. While incorporating more layers or tokens can be beneficial when aggregation is sufficiently large and helps stabilize the representation, intermediate settings tend to introduce noise without sufficient complementary signal, leading to degraded performance.

Effects of Different Observer Models. As shown in Table 5, the performance of S2D depends noticeably on the choice of observer model. Several observers, including *Falcon-7B*, *Qwen2.5-7B*, and *Llama-3.1-8B*, yield near-perfect AUROC and strong TPR at low FPRs, while others, such as *Gemma-2*, perform substantially worse. Importantly, this variation is not explained by model scale, recency, or instruction tuning alone, e.g., the lightweight *OPT-2.7B* remains highly competitive.